

MGMT 682: Digital Product Design

Assignment 1

Please write your answers as clearly and concisely as possible. Please type your answers. Recall the course policy on AI: you are free to use AI tools subject to disclosure when and how it was used and you are fully responsible for everything you turn in. We value coherent and concise responses.

Conjoint analysis: You are helping a startup designing a peloton-inspired rowing machine. The company conducted a conjoint analysis to inform their design process. Below are the results.

Feature	Level	Utils
Monthly Price	\$10	100
Monthly Price	\$20	0
Monthly Price	\$30	-100
Live Classes	Yes	50
Live Classes	No	-50
Number of Different Rivers (e.g. Charles, Loire, Ganges) to Row Down in Video Library	100	-30
Number of Different Rivers (e.g. Charles, Loire, Ganges) to Row Down in Video Library	200	30
Number of Different Rivers (e.g. Charles, Loire, Ganges) to Row Down in Video Library	300	90

1) Calculate the dollar value of live classes and the dollar value of 100 extra videos in the library. You can assume this is linear. (10 points)

Step A: Convert utils to dollars using the price slope (linear):

- From the table: \$10 = 100 utils, \$20 = 0 utils, \$30 = -100 utils.
- A \$10 decrease gives +100 utils, so 1 util = \$0.10 (10 utils = \$1).

Step B: Dollar value of live classes:

- Live Classes Yes = +50 utils, No = -50 utils.
- Difference = 100 utils = \$10.

Step C: Dollar value of +100 videos:

- Videos levels: 100 = -30 utils, 200 = +30 utils, 300 = +90 utils.
- Each +100 videos adds +60 utils.
- +60 utils = \$6 per +100 videos.

Answer to Q1: Live Classes add \$10 of value. Every additional 100 videos adds \$6 of value.

2) What would you advise the startup to prioritize in the design process given it has not yet invested money developing either feature? (10 points)

Based on willingness to pay from the conjoint, live classes (\$10) > +100 videos (\$6).

Recommendation: build live classes first, then expand the video library, unless the build cost or time to launch makes live classes clearly less attractive.

Rationale:

- The conjoint shows a higher willingness to pay for live classes than for adding 100 videos: \$10 for live classes versus \$6 per 100 videos, so live classes deliver greater dollarized customer value per feature unit.
- Prioritizing live classes maximizes initial perceived value and potential revenue lift, only reverse the order if development cost, complexity, or launch timing make live classes inferior on ROI relative to adding videos.
- After launch, validate the impact with pilots and A/B tests to confirm engagement and churn improvements attributable to live classes versus library expansion before scaling resourcing.

3) Suppose you also did the conjoint analysis by customer income segment. You noticed middle-income customers valued the live classes at \$2 while high-income customers valued the live classes at \$30. How would this impact your pricing strategy? (10 points)

Offer tiered pricing with fences: a base plan without live classes for price-sensitive users and a premium plan with live classes priced below the high-income segment's \$30 incremental value, ensuring middle-income users aren't pushed to pay for a feature they value at only \$2.

Hence, the Pricing structure is:

- Base plan: no live classes. Priced for price-sensitive users.
- Premium plan: includes live classes. Price below the high-income (below about \$30 of added value) to drive uptake while keeping the base plan attractive for middle-income users.

Segmentation fences:

- Add fences to limit cross-segment switching, such as priority booking windows, exclusive instructors or series, and community or leaderboard perks tied to live participation in the premium tier
- Keep non-live features largely consistent across tiers, only adding differentiation that reinforces the fence (for example, advanced metrics or badges in premium) to maintain clear self-selection.

Experimentation and adjustments

- Validate willingness to pay with A/B tests on the premium uplift, monitor upgrade rates and churn by segment, then tune the premium differential accordingly.
- Re-assess packaging and price bands if competitors add live classes or discount aggressively, keeping premium-priced below economic value to the customer, and the base plan compelling for price-sensitive users(middle-income group).

Note: I used AI to learn the concepts taught in the class before proceeding with this assignment, and to help with grammar and clarity.